Cross-Mission expiMap Analysis Recovers Established Tissue Responses and Identifies Complementary Pathway Shifts in Mouse Spaceflight Transcriptomes

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Abstract

Spaceflight affects multiple organs, but differences among missions can obscure biological responses that recur across studies. We asked whether expiMap, a pathway-constrained variational autoencoder, could identify reproducible biological programs in mouse spaceflight transcriptomes. NASA Open Science Data Repository bulk RNA-sequencing samples were mapped into tissue-matched expiMap models trained on non-spaceflight ARCHS4 data, with approximately 2,000 highly variable genes linked to current mouse Reactome pathways. Each accession or mission project received equal weight in the summary of flight-ground pathway shifts, which were evaluated using conventional gene-set enrichment, held-out-project prediction, three complete reference-query trainings, cell-composition sensitivity analyses, and member-gene review. Four tissues produced reproducible patterns. In skin, flight was associated with lower chromatin regulation, DNA repair, Hedgehog signaling, sphingolipid metabolism, and cell-junction programs, consistent with reduced tissue maintenance and barrier coordination. Thymus showed lower DNA-repair and cytoskeletal programs, together with a model-specific lower lymphoid-stromal interaction score. Liver showed lower MHC class II antigen-presentation and T-cell receptor scores despite heterogeneous metabolic responses. Spleen showed coordinated lower T-cell receptor signaling, neutrophil degranulation, and C-type lectin receptor signaling across five unconfounded projects and three trainings; all three programs had preranked GSEA false discovery rates below 0.05. The results agree with established tissue responses and motivate tests of skin and thymic tissue maintenance, hepatic immune communication, and splenic innate effector programs. Independent cell-resolved and functional studies are needed to identify which cells drive these pathway shifts and whether they affect biological function.

Keywords: spaceflight; NASA Open Science Data Repository; expiMap; ARCHS4; Reactome; transcriptomics; thymus; skin; liver; spleen

Introduction

Spaceflight combines unloading, radiation, altered circadian and nutritional environments, confinement, and mission-specific operational factors. Its biological effects span immune, integumentary, metabolic, and musculoskeletal systems. NASA's Open Science Data Repository (OSDR) makes molecular and phenotypic data from many missions available through a common discovery framework [1]. OSDR enables cross-mission analyses, although strain, sex, age, mission duration, hardware, tissue collection, and sequencing protocol vary among studies. A pooled flight-ground contrast can therefore be precise while still reflecting the composition of the pooled studies rather than a reproducible spaceflight response.

Most published spaceflight transcriptomic studies interpret lists of differentially expressed genes or enrichments computed independently for each experiment. Those analyses have established several tissue phenotypes. Mouse thymus undergoes atrophy and loss of proliferative programs during flight [6-10]. Spaceflight-exposed skin exhibits barrier disruption, dermal atrophy, oxidative and DNA stress, inflammation, and impaired regeneration [12-16]. Mouse liver develops lipid dysregulation, altered xenobiotic and sulfur metabolism, insulin-signaling changes, and mitochondrial stress [17-23]. Splenic studies report reduced T-cell abundance or activation and altered hematopoietic and innate immune states [34-39]. Kidney studies implicate strain-dependent extracellular-matrix, fibrosis-related, transporter, and nephron responses [40-42]. It is less clear whether an interpretable reference model can recover these responses while revealing additional pathway patterns that recur across studies.

expiMap embeds prior gene-program knowledge directly in the decoder of a conditional variational autoencoder [2]. Each constrained latent dimension represents an annotated gene program, while scArches architecture surgery permits a query dataset to be mapped into a trained reference [3]. The method was developed for single-cell atlases, where latent programs contextualize cell state and perturbation. Here, a large non-spaceflight bulk reference supplies the tissue expression background, and the model maps OSDR samples as whole-tissue profiles. We do not use the latent variables to infer cell types or claim single-cell resolution.

We screened six tissue models built from tissue-matched ARCHS4 references [4], current mouse Reactome programs [5], and approximately 2,000 highly variable genes (HVGs). Detailed biological interpretation focused on thymus, skin, liver, and spleen. Kidney was analyzed as exploratory. Our primary descriptive quantity was the mean project- or accession-specific change in decoder-oriented pathway score, with equal weight assigned to each source. We assessed whether pathway directions agreed with conventional enrichment, predicted held-out mission projects, recurred after complete reference-query retraining, and persisted after adjustment for broad cell-composition proxies. Biological plausibility and reproducibility were evaluated separately. A familiar pathway can be model-sensitive, whereas an internally reproducible expiMap program may lack support from conventional scoring. Statistical tests were treated as complementary evidence rather than a single automatic inclusion threshold.

Materials and methods

Study design

We performed a retrospective, reference-guided analysis of publicly available mouse bulk RNA-sequencing data. The repository screen covered liver, aggregate skeletal muscle, skin, kidney, thymus, spleen, lung, and retina. The muscle-specific screen included soleus because it is an established spaceflight-sensitive antigravity muscle [24-30]. Detailed interpretation was limited to tissues in which at least one manually reviewed pathway showed concordant directions across projects and complete training seeds, was not contradicted by member-gene or tissue-context review, and either passed all five directional checks or remained internally robust despite incomplete support from conventional methods. Thymus, skin, liver, and spleen met these criteria and constitute the primary analyses. Kidney remained exploratory because its related structural and growth-factor pathways lacked conventional FDR support. Soleus did not meet these criteria and is reported only in supplementary screening and sensitivity analyses; no lung or retina pathways met the criteria. Models were trained independently by tissue, so latent-score magnitudes are interpreted within tissues and are not compared numerically across them.

OSDR query discovery and assembly

Studies were discovered through the NASA OSDR Biological Data API rather than a preassembled OSDR HDF5 file [1]. Selection required *Mus musculus*, bulk transcriptomic RNA sequencing, an identifiable spaceflight or ground-control condition, a supported tissue label, and an available unnormalized-count table. Samples not assigned to flight or ground control were excluded. Public processed count tables and ISA-style sample metadata were assembled using the repository pipeline documented in `docs/osdr_api.md`.

The four primary models and the exploratory kidney model mapped 709 OSDR samples. Primary effects were estimated from 700 samples after nine condition-strain-confounded spleen samples were excluded. The four primary tissues included 117 thymus samples (63 flight, 54 ground control; 5 projects), 151 skin samples (80, 71; 6 accessions collapsed to 4 mission projects for project-level checks), 197 liver samples (101, 96; 10 accessions collapsed to 9 projects), and 100 spleen samples (49, 51; 5 projects). The spleen model mapped 109 samples, but OSD-288 was excluded from the primary effect because its recorded strain labels were disjoint by condition. The exploratory kidney model included 135 samples (68, 67; 6 projects). A 53-sample soleus model (28 flight, 25 ground control; 3 projects) was evaluated during screening and is documented only in the supplementary records.

The liver assembly initially contained 231 samples from 12 accessions. OSD-168 was excluded because it repackages RR-1 and RR-3 cohorts represented by the canonical OSD-48 and OSD-137 accessions and includes with-ERCC and without-ERCC technical variants. OSD-164 was excluded because its RR-1 animals overlap OSD-47 and its MiSeq libraries are not an independent biological cohort. The liver query was remapped after both exclusions, preventing technical or library representations of the same animals from being treated as independent studies. Where an ERCC pair must be collapsed elsewhere in the repository, the established policy retains the no-ERCC profile because it is more comparable with the standard matrices used by the other studies; no OSD-168 profile was retained in this primary analysis.

The accession-level sample and covariate audit is provided in Table S2, and the six-model scope is provided in Table S25. Sex was fixed or balanced within each accession's flight-ground comparison. Age was unavailable in the assembled analysis metadata. OSD-289 thymus and OSD-714 soleus had strain labels that were disjoint by condition and were prespecified for restricted sensitivity analysis. OSD-288 spleen was excluded from the primary analysis, rather than treated only in a sensitivity analysis, because its condition-strain disjointness was identified before the spleen pathway review.

ARCHS4 references and Reactome architecture

Tissue-matched, non-spaceflight mouse bulk RNA-sequencing samples were extracted from the ARCHS4 mouse H5 resource [4]. Search terms and metadata filters were tissue specific, and known spaceflight records were excluded. To keep reference size tractable without allowing one GEO series to dominate, series-aware sampling was used where necessary. The references contained 1,362 thymus samples from 107 series, 2,593 skin samples from 185 series, 5,000 liver samples from 518 series, 6,289 spleen samples from 470 series, 2,464 kidney samples from 154 series, and 1,412 skeletal-muscle samples from 88 series. The kidney model used all eligible reference profiles. For spleen, 19 singleton series were excluded only from HVG ranking so that batch-aware selection could run; all 6,289 profiles remained in reference training.

The architecture mask was generated from current official Reactome mouse pathway assignments mapped to mouse Ensembl gene identifiers [5]. For each reference, `scanpy.pp.highly_variable_genes` selected 2,000 genes from raw reference counts with the ARCHS4 condition as the batch key. Genes unavailable after reference-query matching were removed, leaving 1,994, 1,997, 1,995, 1,994, and 1,996 genes for thymus, skin, liver, spleen, and kidney. Pathways with insufficient retained membership were excluded, leaving 387, 319, 364, 360, and 336 constrained latent programs. The soleus screen retained 1,975 genes and 357 programs (Table S25).

expiMap training and query mapping

Reference models used a negative-binomial reconstruction loss, three hidden layers of 300 units, and a Reactome-constrained linear decoder. Reference training was run for up to 400 epochs with early stopping. Primary seed-2020 runs completed 400, 244, 248, 205, and 287 epochs for thymus, skin, liver, spleen, and kidney; the soleus screen completed 400 epochs. OSDR queries were mapped for 250 epochs with accession encoded as the query condition so that mission-specific offsets could be represented during mapping. The de-duplicated liver query was remapped against the unchanged trained liver reference. Posterior-mean latent scores were exported. Training and mapping used an NVIDIA A100-SXM4-40GB GPU. Tables S1, S25, and S26 report the training provenance.

The selected models contain only constrained Reactome dimensions. Here, *complementary* denotes an annotated Reactome program that adds a plausible tissue-specific perspective beyond the dominant phenotype described in prior literature; it does not denote an unconstrained de novo program. De novo variants evaluated during model development were not included because their leading nodes were less stable and frequently recapitulated known pathway content.

Latent mapping and sample-level visualization

Mapping coverage was evaluated separately within each tissue. Reference latent scores were standardized using the ARCHS4 reference mean and variance, a 20-component principal-component model was fitted to the standardized reference, and the corresponding OSDR query scores were projected without refitting. Query-to-reference nearest-neighbor distance was calculated in that 20-dimensional reference PC space. A query was considered within reference support when its nearest-reference distance did not exceed the 95th percentile of leave-self-out reference-to-reference nearest-neighbor distances. This diagnostic tests whether mapped queries occupy the broad reference score support; it is not an external validation of biological accuracy or flight-ground separation.

For sample-level display of one representative retained program per primary tissue, latent signs were decoder-oriented and scores were centered within OSD accession before plotting. Individual samples and paired project means are shown to illustrate the underlying distributions and project dependence. All formal descriptive effects retain equal project or accession weight rather than treating plotted samples as independent replicates.

Pathway direction and primary effect

The sign of an unconstrained latent variable is arbitrary. We oriented every pathway score using `EXPIMAP.latent_directions(method="sum")`, which applies the sign of the summed decoder weights so that positive latent direction corresponds to the model's net upregulation direction. All higher and lower statements in this manuscript use these oriented scores. They describe a latent pathway score and do not imply concordant differential expression of every member gene.

For each pathway and accession, we calculated the mean score in flight minus the mean score in ground control. The descriptive effect was the arithmetic mean of accession-specific differences; when accessions represented paired sites or overlapping sources from one mission project, accession effects were first averaged within project. This prevents a large sample set or multiply represented project from dominating the summary, but it does not make mission projects statistically interchangeable. We also recorded direction counts, pooled-sample Welch and Mann-Whitney tests, Wilcoxon signed-rank tests over accession effects, random-effects estimates, and leave-one-accession-out statistics. Benjamini-Hochberg false-discovery rates (FDRs) were calculated within each tissue where applicable. Because the analysis was exploratory and pathway programs overlap extensively, no single FDR or leave-one-out threshold was used as an automatic biological inclusion rule. Instead, magnitude, direction, project agreement, gene support, metadata quality, and orthogonal method evidence were considered together.

Conventional pathway benchmarks and held-out-project validation

We benchmarked the latent directions against two conventional gene-set analyses on the same samples, HVG universe, and Reactome memberships. First, rank-normalized single-sample GSEA (ssGSEA) was calculated with GSEApy 1.3.0 using weight 0.25 and pathway sizes of 5-500 measured genes [31,32]. Accession- and project-balanced ssGSEA flight-ground effects were then calculated exactly as for expiMap. Second, genes were ranked by their mean project-balanced `log2(CPM + 1)` flight-ground effect, and preranked GSEA was run with 1,000 permutations and weight 1 [31,32]. Agreement was summarized by direction and Spearman rank correlation for all comparable programs, the primary expiMap top decile, the 29 reviewed pathways, and the 37 family representatives. Conventional agreement is triangulation, not a ground-truth test of expiMap superiority.

For internal held-out validation, accession effects from shared mission projects were averaged before validation; this collapses the paired MHU-2 and RR-5 skin sites and the two RR-1 liver accessions into one project effect each. In each fold, one project was withheld, pathway effects were estimated from the remaining projects, and the held-out direction was predicted. The top-decile analysis selected pathways using absolute training-project effect only, without viewing the held-out project. Because every fold still comes from the same public repository, this is internal cross-validation rather than independent external replication.

Full-pipeline training-seed sensitivity

The complete reference and query pipeline was rerun with seeds 2021 and 2022 in addition to the primary seed 2020. Each run retrained the tissue-specific ARCHS4 reference for up to 400 epochs and remapped its OSDR query for 250 epochs with otherwise matched settings. Latent signs were reoriented from each fitted decoder before effects were calculated. Seed-direction support required a program to be active and to have the same direction in all three full-pipeline runs. This is stricter than resampling a fixed trained reference because both reference optimization and query mapping can vary.

Bulk cell-composition proxy sensitivity

Broad composition proxies were derived independently from the Tabula Muris Senis Smart-seq2 mouse atlas [33]. Atlas cell types were consolidated into tissue-relevant broad compartments for thymus, skin, liver, spleen, kidney, and limb muscle. For each compartment, the 30 most specific genes available in the tissue HVG matrix were selected from atlas pseudobulk profiles. Bulk-sample marker scores were calculated as the mean standardized `log2(CPM + 1)` expression of those genes. Marker scores and pathway scores were centered within accession, marker scores were reduced to at most three principal components explaining at least 90% of their variance, and the flight coefficient was estimated with and without those components. Composition support required direction retention and at least 25% retention of the absolute unadjusted coefficient.

This procedure is not cell-type deconvolution. Marker-score variation can represent cell state as well as abundance, composition can mediate a real spaceflight response, and the limb-muscle atlas lacks mature myofibers. The analysis therefore asks whether a pathway direction remains after removal of broad atlas-marker-associated variation; it cannot assign the pathway to a cell type or prove a within-cell effect.

Integrated robustness classification

We summarized robustness using five directional checks: ssGSEA, preranked GSEA, held-out-project concordance of at least two thirds, all-three-seed concordance, and composition-proxy support. The 29-pathway review covered thymus, skin, liver, and soleus. For kidney and spleen, we applied the same checks to every primary top-decile program and then manually assessed member genes, decoder weights, redundancy, and tissue fit. Pathways passing all five checks were classified as *triangulated*. Those passing the held-out, seed, and composition checks but lacking directional support from one or both conventional methods were classified as *internally robust with incomplete conventional support*. Pathways supported in the same direction by both conventional methods and held-out validation, but not by seed or composition sensitivity, were classified as *method-supported but model-sensitive*. All remaining pathways were classified as *sensitivity-dependent*. These descriptive categories are not significance levels and do not require GSEA FDR below 0.05. We report GSEA FDR separately: the spleen findings had statistical support, whereas the kidney axis met the directional checks but remained exploratory. Parameters for these robustness analyses and the training-only held-out selection rule were specified before the corresponding analyses were run, but tissue selection, literature labels, pathway-family review, and protocol subgroups remain retrospective and post hoc.

Confound and gene-level sensitivity analyses

Restricted effects were recomputed after removing OSD-289 from thymus and OSD-714 from soleus. The primary spleen effect excluded OSD-288, with the full 109-sample mapping retained as sensitivity evidence. These removals address recorded condition-strain disjointness; they do not resolve unmeasured differences in age, hardware, or mission duration. The original 12-accession liver mapping was retained as a sensitivity analysis against the 10-cohort primary remap. For secondary gene-level support, OSDR unnormalized counts were transformed to `log2(counts per million + 1)`. Gene-wise pooled Welch tests were adjusted by Benjamini-Hochberg, and accession-specific flight-ground gene effects were averaged with equal accession weight. Liver gene-level results used the de-duplicated primary input. For kidney and spleen, we also compared observed member-gene effects with seed-specific oriented decoder weights. Gene-level results were used to determine whether members of an interpreted pathway supported or contradicted the latent interpretation, not to redefine the expiMap score.

Protocol-context sensitivity

The broad OSDR flight label can combine biologically distinct comparisons. We therefore performed post hoc, within-accession context contrasts when sample names and official study metadata supported them (Table S8; Fig. 5). We refer to this procedure as *depooling*: the trained model and each sample's pathway score remain fixed, but a pooled flight-minus-ground contrast is replaced by matched contrasts among protocol-defined sample groups. Depooling does not add samples, retrain expiMap, or make the subgroup comparisons independent. It tests whether an aggregate effect is shared across contexts or instead reflects cancellation among opposing anatomical-site, gravity, recovery, strain, duration, or collection effects.

MHU-2 dorsal and femoral skin were separated into microgravity and onboard centrifuge-generated artificial 1 g groups, each compared with the same Earth 1 g controls. RR-5 dorsal and femoral skin represented approximately 30 days of terrestrial recovery after flight. Paired animal identifiers confirmed that the MHU-2 dorsal and femoral accessions and the RR-5 dorsal and femoral accessions were distinct anatomical samples from shared experimental cohorts, not independent missions or duplicated transcriptomes. We therefore averaged paired-site accession effects within MHU-2 and RR-5 before averaging the four skin projects MHU-2, RR-5, RR-6, and RR-7. RR-6 dorsal skin was separated into approximately 30-day live-return and approximately 60-day ISS-terminal collections. RR-7 dorsal skin was separated by strain and 25-

versus 75-day terminal collection. These contrasts characterize heterogeneity; because they were not prespecified and some share controls, they are not independent confirmatory tests. Collection endpoint was also considered when interpreting thymus and soleus accession effects.

Literature review and evidence labels

Primary spaceflight literature available through July 10, 2026 was reviewed for each tissue. Literature-aligned pathways recovered an established tissue phenotype with a compatible direction. Complementary pathways suggested an additional biological process but lacked direct prior validation in the same tissue or condition. Context-sensitive pathways changed or materially attenuated in a restricted analysis, showed opposing directions across studies, conflicted with tissue-specific prior evidence, or belonged to an overlapping family whose nested latent nodes opposed one another. Protocol stratification did not automatically change these literature roles. Tables 2, S24, and S27-S30 report numerical evidence and review provenance separately from the biological role.

To avoid limiting interpretation to the pathways displayed in the main figures, we also performed broader rule-based reviews. For thymus, skin, liver, and soleus, candidates included the top 10% of active programs by absolute study-balanced effect, directionally stable programs through rank 40, and every prespecified display pathway. This yielded 153 records, which were consolidated into 37 process families using Reactome hierarchy, gene-set overlap, tissue context, and primary literature. For kidney and spleen, every primary top-decile program was manually reviewed, including off-tissue labels and broad parent programs, before three pathways per tissue were retained. Pathways were flagged when overlapping or nested terms made an apparently distinct mechanism redundant or directionally unresolved. Tables S3, S10-S11, and S27-S29 contain the complete and intermediate results.

Results

Model scope and reference coverage

The five models mapped 709 OSDR samples to 17,708 tissue-matched ARCHS4 profiles across 1,434 GEO series (Table 1; Fig. 1). The primary effect analyses used 700 samples after OSD-288 spleen was excluded; the four primary tissues contributed 565 samples. HVG and pathway-membership filtering retained 1,994-1,997 genes and 319-387 Reactome programs per model. Each model therefore contained hundreds of interpretable programs within a similarly sized gene space.

In the four primary tissue models, every mapped OSDR query fell within the tissue reference's 95th-percentile nearest-neighbor distance in 20-dimensional reference PC space (Fig. 2; Tables S31-S32). The first two reference PCs retained visible study structure, and flight and ground samples overlapped rather than forming a universal condition cluster. Project-centroid segments and project-centered representative-program distributions show the within-project shifts underlying the balanced pathway effects (Fig. S9; Table S33). The plots document reference coverage and sample-level distributions; formal results use the project-aware comparisons below.

Table 1. Model scope.

Tissue	Analysis role	ARCHS4 reference	Mapped OSDR	Primary effect samples	Flight	Ground	Primary projects	Genes	Programs
Thymus	Primary	1,362	117	117	63	54	5	1,994	387
Skin	Primary	2,593	151	151	80	71	4	1,997	319
Liver	Primary	5,000	197	197	101	96	9	1,995	364
Spleen	Primary	6,289	109	100	49	51	5	1,994	360
Kidney	Secondary exploratory	2,464	135	135	68	67	6	1,996	336

Supplementary source tables report all 1,427 pathways from the thymus, skin, liver, and soleus screen and the 153-pathway expanded family review (Tables S3 and S9-S11; Figs. S1-S3). The kidney and spleen screens report every pathway in Table S27 and every manually reviewed top-decile program in Table S28. Off-tissue labels, broad parent terms, seed reversals, nested terms, and latent scores contradicted by member genes were not selected on magnitude alone. Spleen met the primary criteria; kidney remained exploratory. Tissue models that did not meet the interpretation criteria are reported in the supplement.

Directional robustness and statistical support differ

Among primary-run top-decile programs for the retained thymus, skin, and liver models, ssGSEA and preranked GSEA direction agreement was 86% and 86%, 77% and 87%, and 83% and 91%, respectively (Fig. S4; Tables S12-S14). Held-out-project direction prediction was 99%, 75%, and 75%. Complete retraining further narrowed the reproducible pathway set: 71% of reviewed thymus, 71% of skin, and 67% of liver programs retained one direction across all three runs. Corresponding results for the soleus screen are reported in Figures S4-S8 and the supplementary source tables.

The spleen and kidney models used the same five directional checks, supplemented by member-gene and decoder-weight review (Figs. 3, 4; Tables S27-S29 and S34-S35). Three spleen programs passed all five checks, were lower in all five unconfounded projects, and had preranked-GSEA FDR below 0.05. Three kidney programs also passed all five directional checks, but their GSEA FDR values were 0.156-1.000, and composition adjustment retained only 29%-36% of their unadjusted magnitude. The kidney directions were repeatable, but their weak GSEA support and attenuation after composition adjustment limit the biological claim.

Table 2. Retained pathway-level results. Median shifts are across three complete reference-query trainings.

Tissue	Retained programs	Median flight-minus-ground shifts	Project direction	GSEA FDR	Interpretation
Thymus			5/5 lower for each	<0.001; 0.154; 1.000	First two triangulated; niche interaction internally

Tissue	Retained programs	Median flight-minus-ground shifts	Project direction	GSEA FDR	Interpretation
	DNA repair; RHOA cytoskeletal cycle; lymphoid-stromal interactions	-0.902; -0.478; -0.617			robust but conventionally discordant
Skin	Chromatin regulation; DNA repair; Hedgehog; sphingolipid metabolism; cell-cell junction organization	-0.701; -0.257; -0.364; -0.194; -0.162	3/4 lower for each	<0.001; <0.001; 0.280; 0.897; 0.007	Four triangulated directions; junction result lacks ssGSEA support
Liver	MHC class II antigen presentation; T-cell receptor signaling	-0.158; -0.064	8/9 lower for each	0.121; 0.051	Directionally triangulated lower adaptive-immune pattern
Spleen	T-cell receptor; neutrophil degranulation; C-type lectin receptor signaling	-0.216; -0.182; -0.105	5/5 lower for each	0.042; <0.001; 0.017	Main adaptive result with complementary innate-effector programs
Kidney	ECM proteoglycans; WNT; IGF transport and uptake	+0.096; +0.067; +0.077	5/6; 6/6; 5/6 higher	0.156; 1.000; 0.433	Secondary exploratory structural and growth-factor pattern

Thymus shows lower repair and cytoskeletal programs

The primary thymus model recovered the established involution phenotype: mitotic cell-cycle and T-cell receptor scores were lower, consistent with reduced thymocyte proliferation and adaptive activity [6-9]. Individual latent programs were less stable than this family-level pattern. Cell-cycle direction changed in one complete training, and the T-cell receptor coefficient approached zero after composition adjustment. We therefore did not use either score as a central expiMap result.

DNA repair and the RHOA cytoskeletal cycle were lower in all five projects, agreed with both conventional methods, predicted every held-out-project direction, retained direction across all trainings, and retained 59% and 82% of their effects after composition adjustment (Figs. 3, 4). Lower lymphoid-stromal interaction also persisted across projects, trainings, and composition adjustment, but both conventional methods disagreed. The reproducible thymus pattern comprises lower repair and cytoskeletal scores and a possible lower thymocyte-niche interaction that requires cell-resolved validation.

Higher innate Toll-like receptor and extracellular-matrix scores in the primary run were not reproducible. The Toll-like receptor effect approached zero and reversed in both additional trainings, while composition adjustment reduced the matrix coefficient to 14% of its original magnitude. Prior evidence still makes innate and stromal responses biologically relevant [10,11], but the current model does not support a stable adaptive-to-innate switch.

Skin shows lower regulatory, repair, and barrier-maintenance programs

Prior skin studies report inflammation, DNA and oxidative stress, barrier disruption, collagen turnover, and impaired regeneration [12-16]. Four lower programs passed every directional check: chromatin-modifying enzymes, DNA repair, Hedgehog signaling, and sphingolipid metabolism. Each recurred across all trainings, predicted 3 of 4 held-out projects, and retained 69%-85% of its effect after composition adjustment (Figs. 3, 4). Cell-cell junction organization was also lower in 3 of 4 projects and stable across trainings and composition adjustment; preranked GSEA agreed, whereas ssGSEA did not. The combined pattern describes a lower regulatory, damage-response, regenerative, barrier-lipid, and intercellular-coordination state rather than one isolated wound-healing pathway.

Keratinization remained a strong primary-model phenotype but not a stable latent node. Its seed effects were -1.22, -2.16, and +0.25. Both conventional methods, most projects, and published cross-mission work supported a lower barrier-differentiation state [12]. This evidence supports its biological plausibility, but keratinization was excluded from the reproducible expiMap result because the node reversed under complete retraining.

Protocol stratification of the pooled groups clarified the apparent mission heterogeneity (Fig. 5; Table S8). Microgravity scores were lower than Earth 1 g for all eight protocol-stratified programs at both MHU-2 skin sites. Onboard artificial 1 g scores were also lower in dorsal skin but higher for most femoral-skin programs, including keratinization, so the data do not support a general artificial-gravity rescue. RR-5, sampled after terrestrial recovery, was the main project-level exception. RR-6 showed the same direction in live-return and terminal collections but a larger effect in the longer terminal group; RR-7 effects depended on strain and duration. Averaging paired MHU-2 and RR-5 sites within each project preserved lower scores for all eight displayed programs, and the six-accession and four-project effects correlated at r=0.995 across all active skin programs. Paired-site weighting did not create the lower skin pattern. The stratified analyses instead show where that pattern weakens or reverses.

Liver shows lower adaptive-immune programs amid heterogeneous metabolism

Mouse spaceflight studies consistently implicate hepatic lipid handling, xenobiotic metabolism, sulfur and glutathione biology, insulin signaling, mitochondrial stress, and tissue remodeling [17-23]. The primary model reproduced involvement of these families without a uniform direction. The cytochrome P450 score was higher on average, but accession-specific effects were positive in 5 accessions and negative in 5; insulin-regulatory and glutathione nodes changed direction or became inactive in one training. The split among accessions is consistent with differences among missions, strains, and enzyme families rather than a single metabolic switch.

MHC class II antigen presentation and T-cell receptor signaling formed the most reproducible non-metabolic pattern. Both were lower in 8 of 9 projects, agreed in direction with both conventional methods, recurred in all trainings, and retained 53% and 104% of their effects after composition adjustment (Figs. 3, 4). Broader class-I processing, neutrophil, complement, and interleukin terms were also predominantly lower in the expanded review, but they were treated as correlated members of one immune-effector family rather than as separate discoveries. Bulk data cannot distinguish lower immune-cell abundance from altered signaling within resident or infiltrating cells.

Rho-family and extracellular-matrix programs were not directionally stable. Rho-family direction changed across trainings. Matrix organization was predominantly lower in the primary run and agreed with conventional methods, but crossed zero after composition adjustment. The matrix family was involved in the hepatic response, but these data do not establish a composition-independent direction.

Spleen shows lower adaptive and innate immune programs

The spleen model excluded OSD-288 from the primary effect because strain was disjoint by condition. T-cell receptor signaling was lower in all five retained projects and all three trainings, with a median shift of -0.216 and GSEA FDR 0.042 (Figs. 3, 4). This agrees with reports of reduced splenic T-cell abundance, activation, and anti-CD3/CD28 responsiveness after spaceflight [35-37]. Sixty-two percent of measured member genes had lower expression, and a median 69% of absolute decoder weight predicted the observed gene direction.

Neutrophil degranulation and C-type lectin receptor signaling were also lower. Neutrophil degranulation had a median shift of -0.182 and GSEA FDR below 0.001; C-type lectin receptor signaling had a median shift of -0.105 and GSEA FDR 0.017. Both were lower in all five projects, all trainings, ssGSEA, preranked GSEA, and composition-adjusted analyses. Approximately 62%-63% of measured member genes and 76%-78% of median absolute decoder weight supported the directions. Across projects, the three scores moved together: T-cell activation, pathogen sensing, and degranulation-related transcription were all lower in flight.

The degranulation score is not a functional assay. Human postflight neutrophils can show activated, T-cell-suppressive phenotypes [39], and a lower whole-spleen transcriptomic program could reflect cell abundance, maturation, localization, or altered state. These findings therefore support a lower splenic degranulation-related transcriptional program, but not reduced neutrophil function. CD28-linked activation and lymphoid-to-nonlymphoid interaction were also lower across projects and trainings but disagreed with preranked GSEA; MHC class II antigen presentation was lower in only 3 of 5 projects. These terms provide context but do not define the central result.

Kidney shows an exploratory structural and growth-factor pattern

The exploratory kidney result comprised three higher programs: ECM proteoglycans, aggregate WNT signaling, and IGF transport and uptake (Fig. 4). Their median shifts were +0.096, +0.067, and +0.077. WNT was higher in all six projects, and ECM proteoglycans and IGF transport were higher in five. All retained their direction across three trainings and after composition adjustment. Eighty-one to 86% of measured member genes shared the pathway direction, while a median 63%-80% of absolute decoder weight predicted the observed member-gene directions.

Previous renal studies have reported ECM dysregulation, fibrosis-related signaling, nephron remodeling, and WNT involvement [40,41]. Here, matrix structure, WNT signaling, and IGF availability moved together, consistent with a possible repair or maladaptive-growth state. The scores do not establish fibrosis, growth-factor protein activity, or a uniform response of every WNT ligand.

The evidence remains exploratory. Preranked-GSEA FDR was 0.156 for ECM proteoglycans, 1.000 for WNT, and 0.433 for IGF transport, and composition adjustment retained only 29%-36% of the unadjusted effects. Lower amino-acid metabolism was rejected because only 27% of measured genes shared the latent direction, decoder-weighted concordance was 42%, and ssGSEA, GSEA, and prior kidney analysis pointed higher [42]. The *Biological oxidations* program was interpreted as heterogeneous CYP, UGT, and ACSM regulation rather than a broad decrease. The numerically repeatable metabolic labels therefore do not support a single lower kidney metabolic response.

Pathway patterns differ among tissues

Each tissue had a distinct pattern. Thymus showed lower repair and cytoskeletal scores and a possible niche-interaction component. Skin combined lower regulatory, repair, regenerative, barrier-lipid, and junction scores. In liver, lower adaptive-immune communication occurred alongside heterogeneous metabolic effects. Spleen showed lower adaptive activation, innate sensing, and effector scores. Kidney alone showed higher matrix and growth-factor scores, although the evidence was weaker. No direction or process was shared strongly enough to support a universal spaceflight pathway.

These programs are not statistically independent. Reactome terms overlap, and even latent scores with little gene overlap can covary because of cell composition, model structure, or a shared condition axis. We therefore interpret correlated programs as nonredundant tissue-state hypotheses rather than as separate mechanisms.

Discussion

This study applies an interpretable reference-mapping architecture to a use case for which it was not designed: cross-mission bulk tissue. The primary models recovered major elements of prior literature, but the robustness analyses show that biological familiarity is insufficient evidence of latent-program reproducibility. Top expiMap pathways agreed with conventional enrichment substantially more often than the full latent background, while complete reference-query retraining changed or inactivated several prominent programs. expiMap organizes the results biologically, but this analysis does not show that it outperforms conventional enrichment, and several named latent nodes were unstable.

Biological interpretation after robustness filtering

After robustness filtering, four primary tissue patterns remained, with kidney as an exploratory extension (Figs. 3-6). Figure 6 distinguishes three evidentiary layers: established phenotypes from prior literature, lower program scores observed in this study, and complementary tissue-state hypotheses prompted by those observations. Its dotted links represent inference, not a causal pathway sequence. In skin, established barrier injury and inflammation were accompanied by lower regulatory, repair, regenerative, barrier-lipid, and cell-junction scores. This pathway family was coherent even though individual terms differed in conventional FDR and the junction result lacked ssGSEA support. The results point to a lower tissue-maintenance state whose magnitude depends on gravity, anatomical site, recovery, strain, and duration. This is more specific than a general change in wound healing.

The primary thymus run suggested a broad shift from adaptive to innate programs. Robustness analyses narrowed this interpretation to lower repair and cytoskeletal programs with a possible lower lymphoid-stromal interaction. The higher innate and matrix nodes were not stable after complete retraining or composition adjustment. The supported thymus interpretation adds repair, cytoskeletal, and niche programs to known involution without claiming an innate switch.

Liver results differed by pathway family. Established metabolic families were repeatedly involved but did not share a uniform direction across missions. MHC class II antigen presentation and T-cell receptor signaling formed a more reproducible lower immune-communication pattern. These immune scores extend the predominantly metabolic literature, although bulk data cannot resolve cell abundance from cell state.

Spleen provided the most consistently supported multi-pathway finding. Lower T-cell receptor signaling recovered prior evidence of adaptive impairment, while lower C-type lectin receptor and neutrophil degranulation programs extended that phenotype to innate sensing and effector transcription. All three were lower across unconfounded projects, trainings, conventional methods, and composition-adjusted analyses and had GSEA FDR below 0.05. Because these are whole-spleen transcriptomic scores, they may reflect immune-cell abundance, localization, maturation, or altered regulatory state; they do not establish neutrophil inactivity.

Kidney ECM proteoglycan, WNT, and IGF-transport scores rose together, linking structural and growth-factor responses previously implicated separately in renal spaceflight biology. Their project, seed, member-gene, and decoder support justify follow-up, but high GSEA FDR and composition attenuation prevent an equal-strength claim. We therefore report directional reproducibility separately from statistical support.

Disagreement with one conventional score did not always coincide with instability. Thymic lymphoid-stromal interaction and skin cell-cell junction organization remained reproducible within the latent framework and may capture weighted gene relationships that rank-based methods miss. By contrast, agreement with prior literature did not guarantee model stability, as shown by skin keratinization and thymic cell cycle. External datasets should test the reported pathway directions directly, rather than selecting a new set of biologically appealing pathways after each analysis.

Strengths and limitations

The analysis used API-based OSDR provenance, tissue-matched references, current mouse Reactome mappings, decoder-oriented latent signs, and project-aware weighting. It reports every pathway result rather than only selected findings. Additional checks included same-gene conventional benchmarks, training-only held-out-project prediction, complete reference-query seed reruns, member-gene and decoder review, and a single-cell-atlas marker sensitivity. The tissue-selection audit documents why each tissue was included or excluded. Failed checks are reported alongside positive results so that agreement with prior literature does not substitute for reproducibility.

Several limitations concern validation and model scope. There is no independent external flight cohort: project-wise validation reuses the same repository and cannot estimate performance in a new mission. expiMap was developed for single-cell reference mapping, and bulk latent shifts can reflect either cell abundance or cell state. The Tabula Muris Senis analysis provides broad marker proxies rather than deconvolution. Three training seeds reveal some optimization sensitivity but do not characterize every source of model uncertainty, and independently trained scales cannot be compared quantitatively across tissues.

Feature selection and retrospective interpretation introduce further uncertainty. HVG selection omits low-variance genes, while overlapping Reactome nodes can partition shared genes in opposite directions. Tissue and pathway selection, family consolidation, literature labels, and protocol subgroups were retrospective and post hoc; the directional robustness labels do not retroactively prespecify discovery. Age was unavailable, and mission, endpoint, strain, sex, vendor, duration, and hardware remain partly inseparable.

Some limitations are specific to the assembled cohorts. Paired skin sites are not independent missions, spleen excludes one confounded project, and unrecognized overlap may persist in public assemblies. Kidney passed all five directional checks, but all three GSEA FDR values exceeded 0.05. A latent score neither implies that every member gene changes together nor establishes causal pathway activity. The reported directions are candidates for external, cell-resolved validation, not definitive biomarkers.

Conclusion

Reference-guided expiMap produced interpretable pathway scores across heterogeneous OSDR datasets, but only a subset remained consistent across project, method, seed, composition, and member-gene checks. Skin showed lower regulatory, repair, regenerative, barrier-lipid, and junction scores. Thymus showed lower repair and cytoskeletal programs together with a model-specific lower lymphoid-stromal interaction. Liver showed lower adaptive-immune scores alongside heterogeneous metabolic responses. Spleen provided the most consistently supported multi-pathway result: lower T-cell receptor, neutrophil degranulation, and C-type lectin receptor programs across all unconfounded projects and orthogonal checks. Kidney yielded an exploratory higher matrix and growth-factor pattern that requires stronger conventional and composition-independent support. These directions are targets for independent single-cell, spatial, biochemical, and functional validation; they do not constitute a causal model or a universal spaceflight signature.

Data and code availability

All source data are publicly available from the NASA Open Science Data Repository, ARCHS4, Reactome, and the Tabula Muris Senis CELLxGENE distribution. The code repository is https://github.com/jasont314/nasa-mouse. OSDR discovery and file-access documentation is in `docs/osdr_api.md`. The official mouse Reactome-derived GMT is `data/pathways/reactome_current_mouse_ensembl.gmt`. Exact model directories, source tables, and rebuild commands are listed in `paper/asgsr_expimap_hvg/supplementary_methods.md`. Analysis scripts are under `src/nasa_mouse_expimap/`. `build_asgsr_paper.py` generates the base figures and tables. Conventional, held-out, composition, and seed analyses use `reviewer_robustness_analysis.py`, `run_asgsr_seed_sensitivity.py`, and `integrate_reviewer_robustness.py`. Kidney and spleen analyses use `run_kidney_spleen_seed_sensitivity.py`, `analyze_kidney_spleen_reassessment.py`, and `curate_kidney_spleen_reassessment.py`; `integrate_reassessed_tissues_paper.py` incorporates those results, and `build_publication_figures.py` produces publication-scale figures and their source tables. Annotation prompts, accepted rationales, and source links are recorded in `docs/annotation_prompts.md` and `docs/annotation_provenance.md`. No archival DOI has been assigned.

Author contributions

Jason Trinh: conceptualization, methodology, software, formal analysis, investigation, visualization, writing, and project administration. Confirm contributor roles before submission.

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Competing interests

The author declares no competing interests. Confirm this statement before submission.

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Figure captions

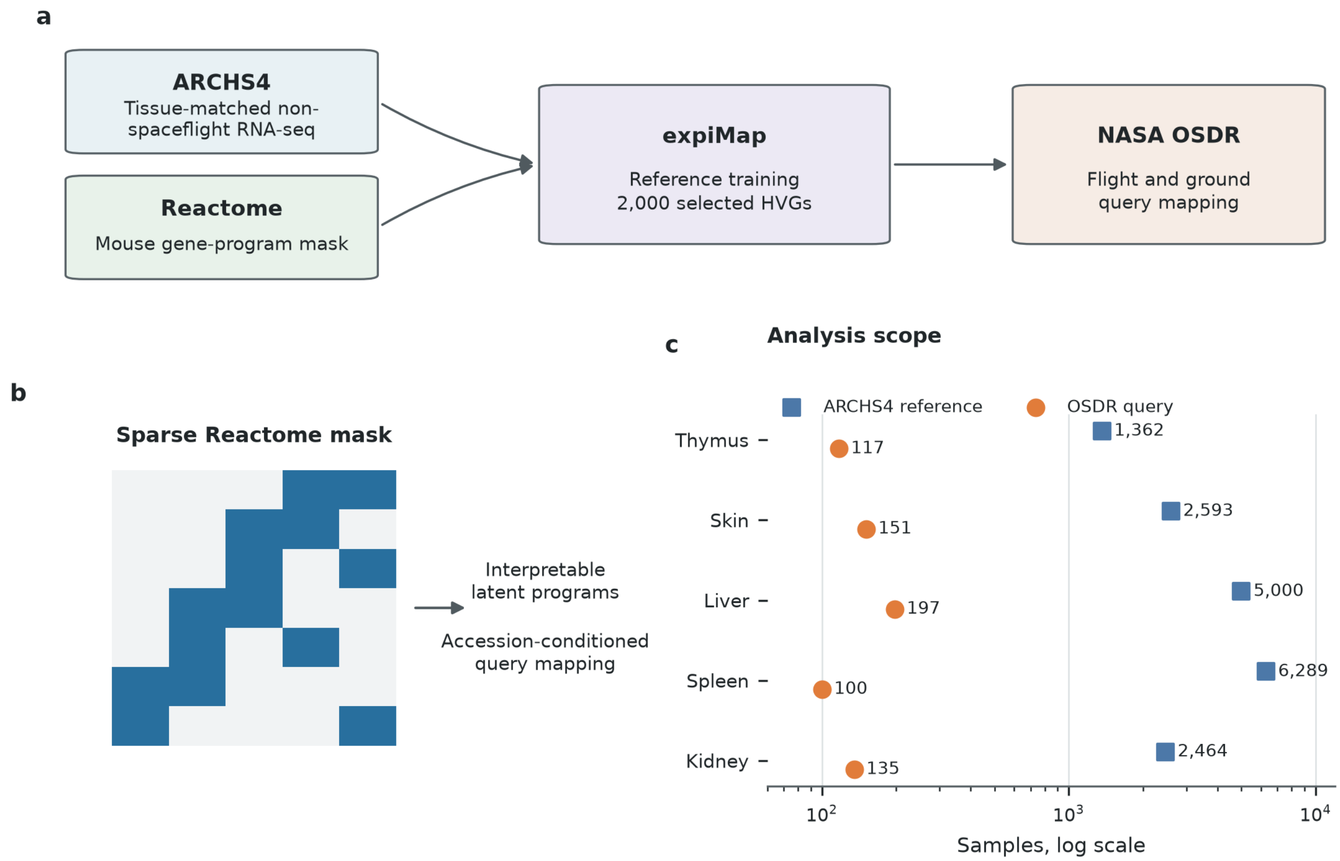


Figure 1. Reference-guided workflow, constrained architecture, and analysis scope. a, Tissue-matched non-spaceflight ARCHS4 profiles and current mouse Reactome memberships define each reference model; NASA OSDR flight and ground profiles are then mapped as accession-conditioned queries. **b,** The sparse architecture connects measured genes to annotated latent programs. **c,** Reference and primary-query sample counts for the four primary tissues and the exploratory kidney model. Spleen shows the 100 samples retained for its primary effect after OSD-288 was excluded.

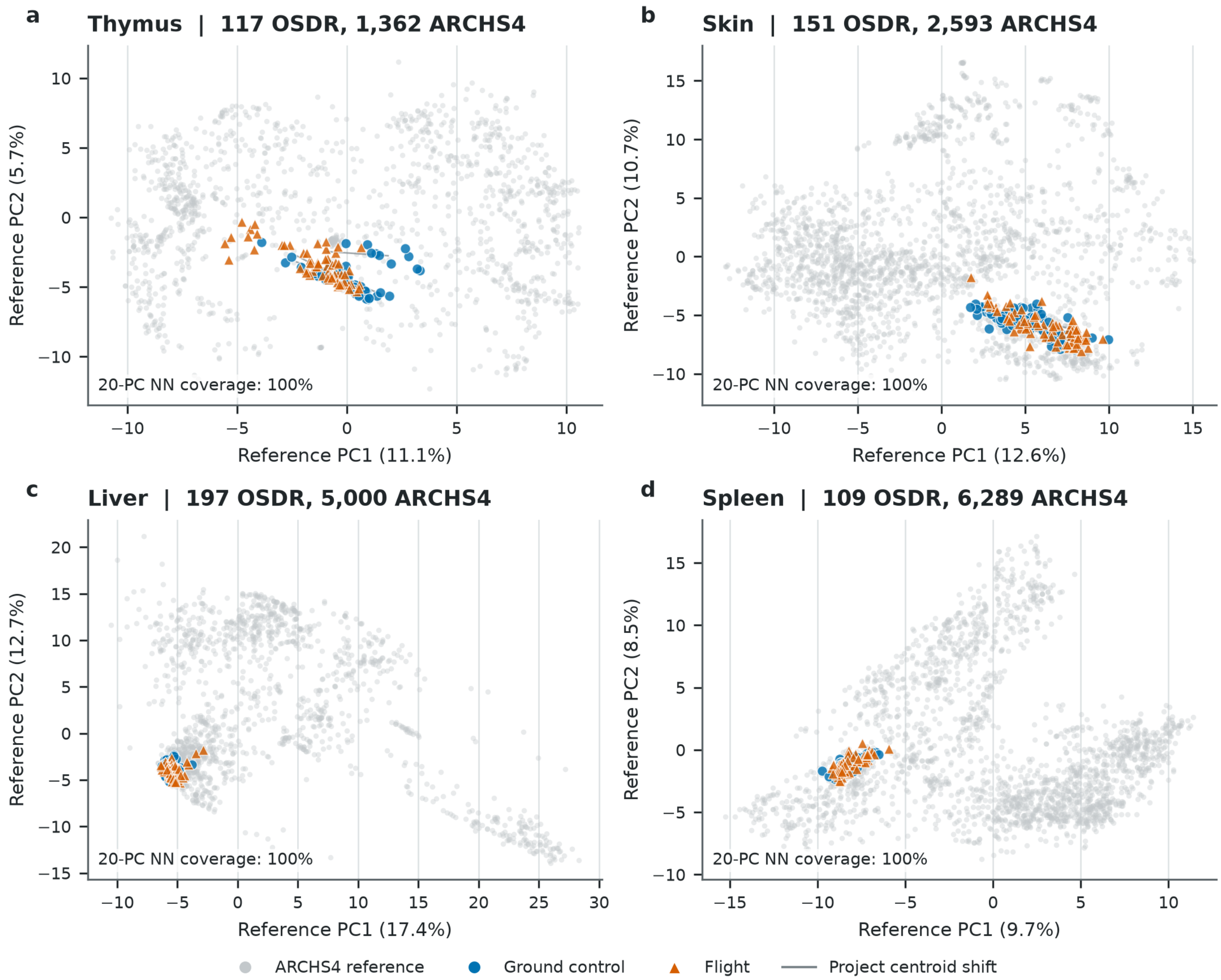


Figure 2. OSDR queries map within tissue-matched ARCHS4 latent-program space. Principal components were fitted separately to each standardized ARCHS4 reference and applied to its OSDR query without refitting. Gray points are reference profiles; blue circles are ground controls and orange triangles are flight samples. Thin segments connect ground and flight centroids within an OSD project. The displayed nearest-neighbor coverage is the fraction of query samples whose distance in 20-dimensional reference PC space is within the 95th percentile of leave-self-out reference distances. Coverage is a mapping diagnostic, not evidence of condition separation or external biological validation.

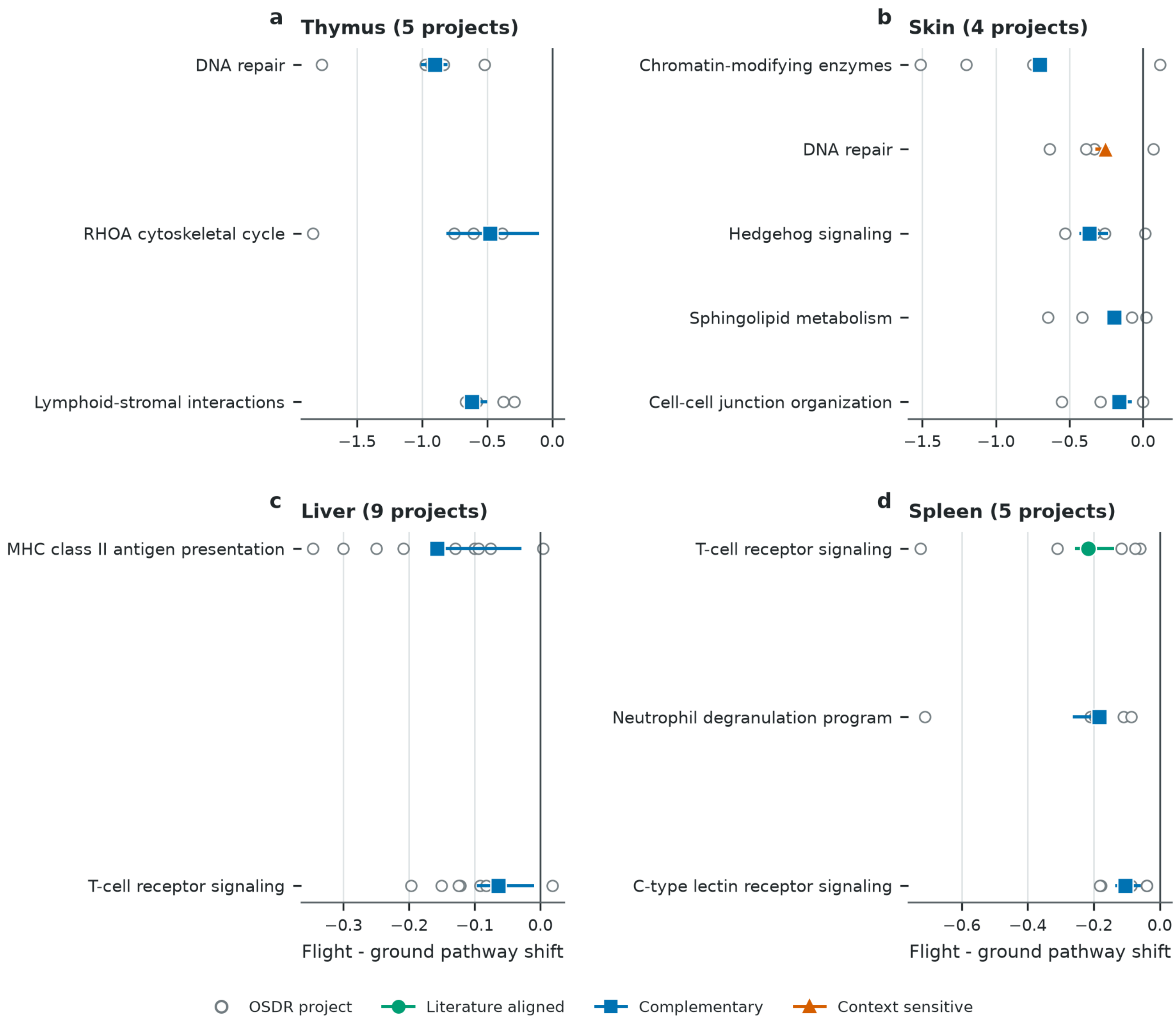


Figure 3. Retained pathway directions across projects and complete training runs. Open circles are primary-seed project effects. Colored segments span three complete reference-query trainings, and the marker is their median. Circle, square, and triangle markers redundantly encode literature-aligned, complementary, and context-sensitive roles. All displayed scores from the primary tissues are lower in flight. Spleen excludes OSD-288 from the primary project effect. Effects are decoder-oriented, and axes are tissue specific.

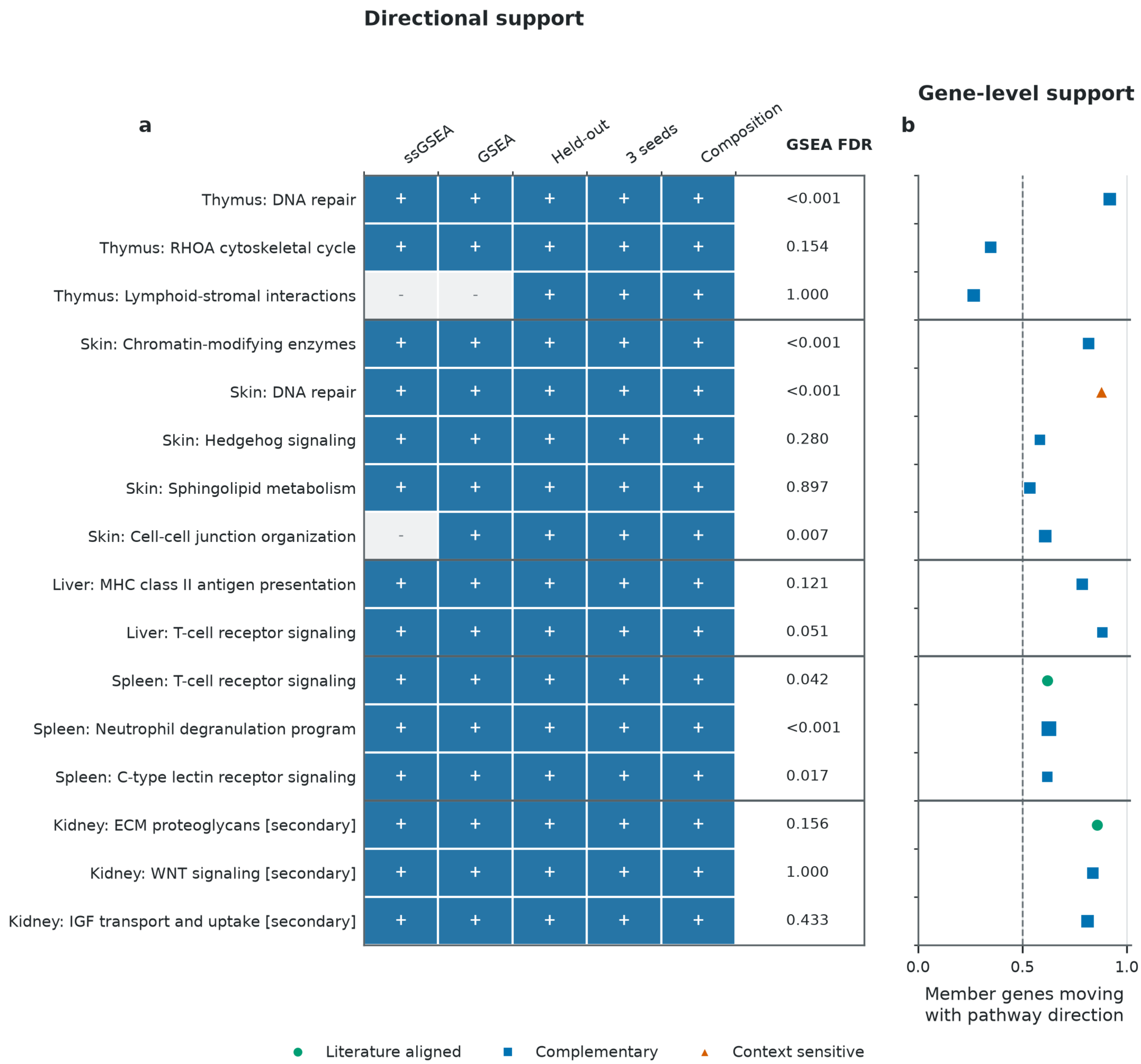


Figure 4. Orthogonal directional checks and member-gene support for retained pathways. a, Blue cells and plus symbols indicate direction agreement with ssGSEA, preranked GSEA, held-out projects, all three complete trainings, or composition-proxy adjustment; pale cells and minus symbols indicate a failed directional check. GSEA FDR is reported separately because the five-check status is not a significance threshold. **b,** Points show the fraction of measured Reactome member genes whose project-balanced expression effect shares the seed-2020 pathway direction; the dashed line marks 50%. Marker shape and color encode the literature role as in Figure 3, and point size reflects measured member-gene count. Kidney is explicitly secondary because its directions are composition-attenuated and lack conventional FDR support.

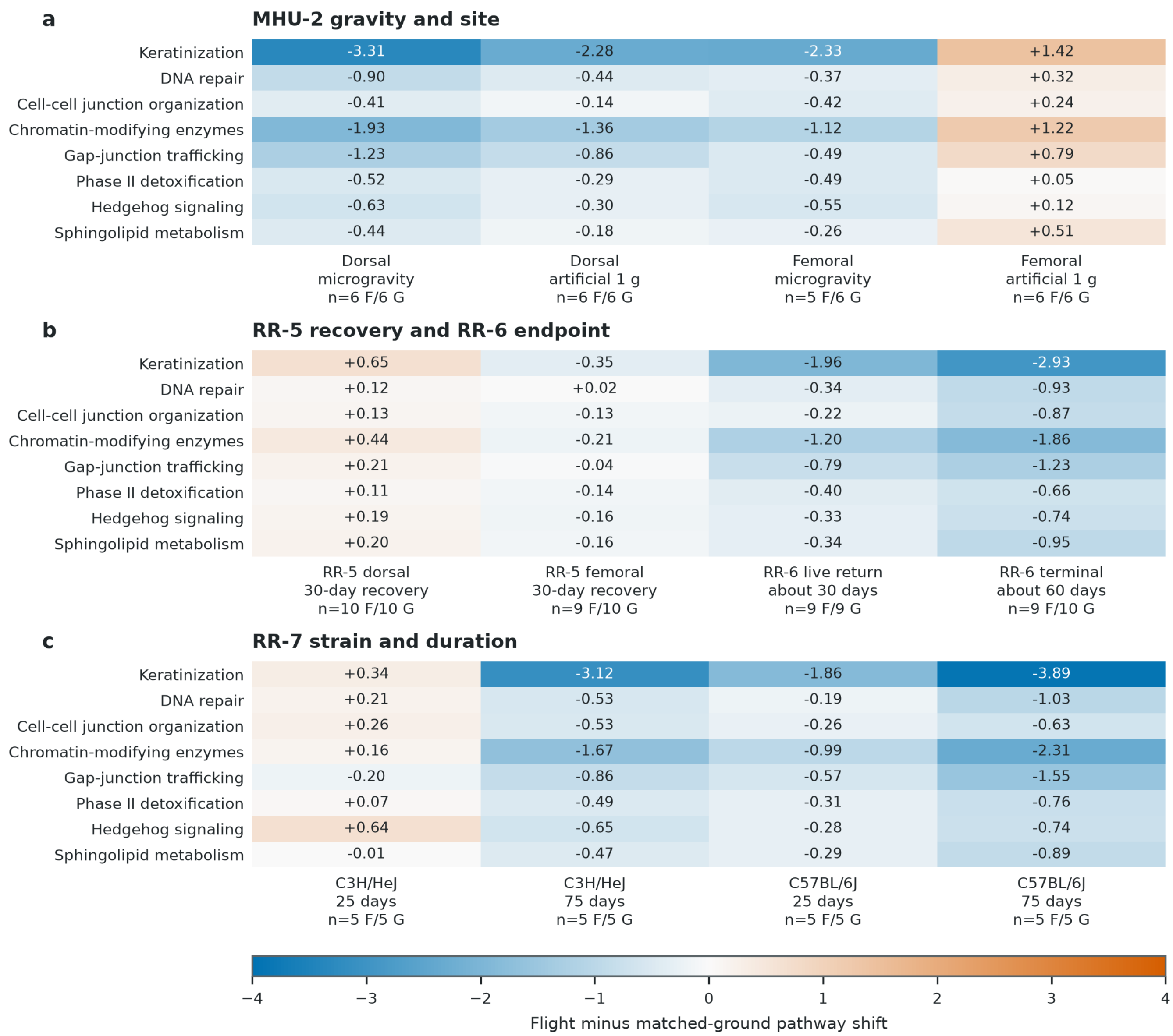


Figure 5. Depooled skin pathway effects by protocol context. Cell values are decoder-oriented flight-minus-matched-ground scores calculated from fixed sample scores, with exact flight and ground sample counts under each contrast. **a,** MHU-2 microgravity and onboard artificial 1 g subgroups share Earth 1 g controls and are separated by dorsal and femoral site. **b,** RR-5 was collected after terrestrial recovery, whereas RR-6 separates live-return and ISS-terminal collection. **c,** RR-7 separates strain and duration. Paired sites and shared controls are not independent experiments. These post hoc contrasts show cancellation and context dependence but are not independent confirmatory tests.

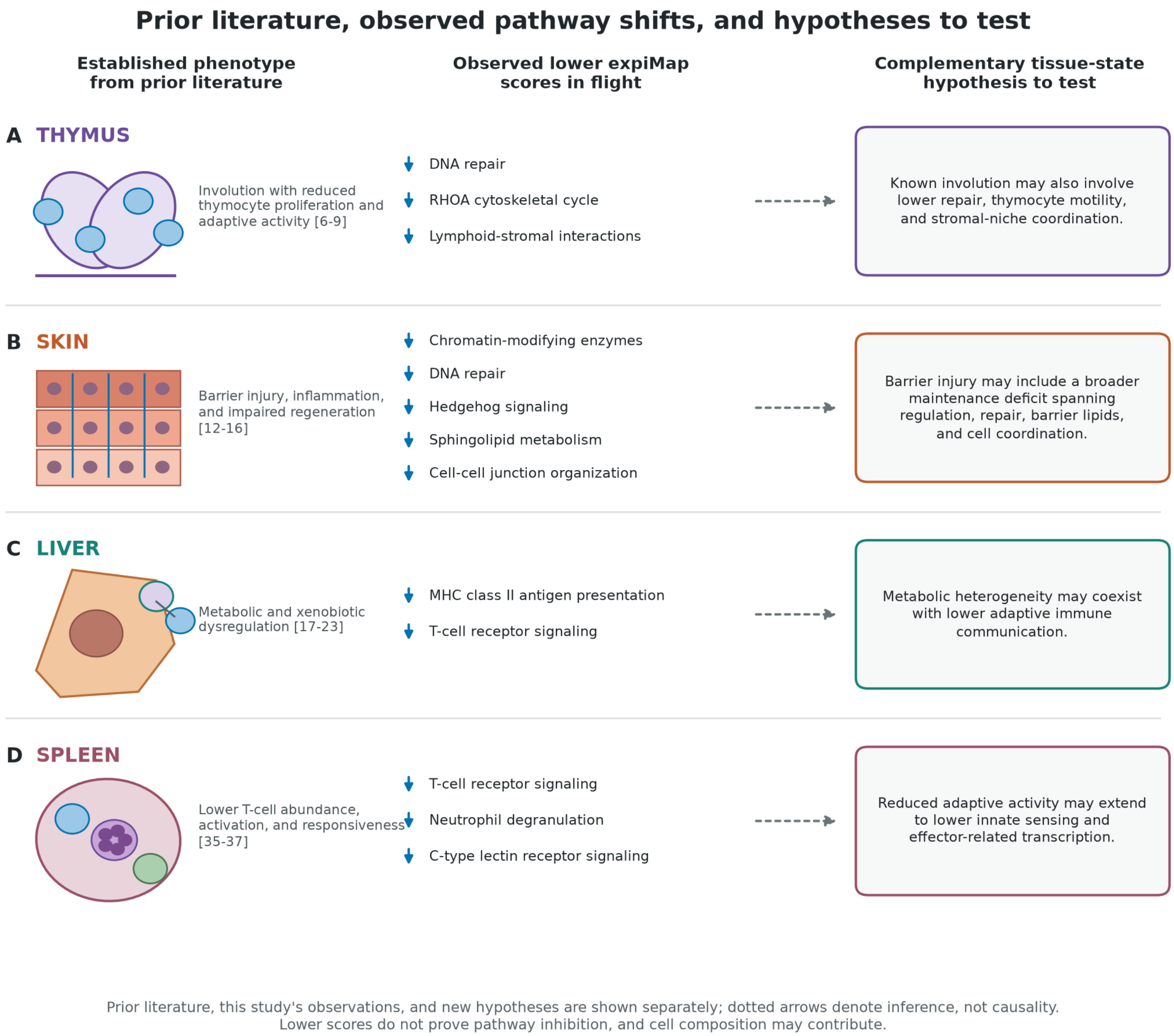


Figure 6. Prior literature, observed pathway-score directions, and complementary tissue-state hypotheses. The left column summarizes established tissue phenotypes from the cited literature, the middle column lists retained lower flight scores observed in this analysis, and the right column states new hypotheses motivated by integrating those observations with prior context. Dotted arrows denote inference, not mechanistic or causal order. The illustrations do not assign bulk-tissue signals to specific cell types. Lower expiMap scores do not prove biochemical pathway inhibition, and cell-composition changes may contribute. The schematic summarizes results from Figures 3-4 and contains no additional analysis.

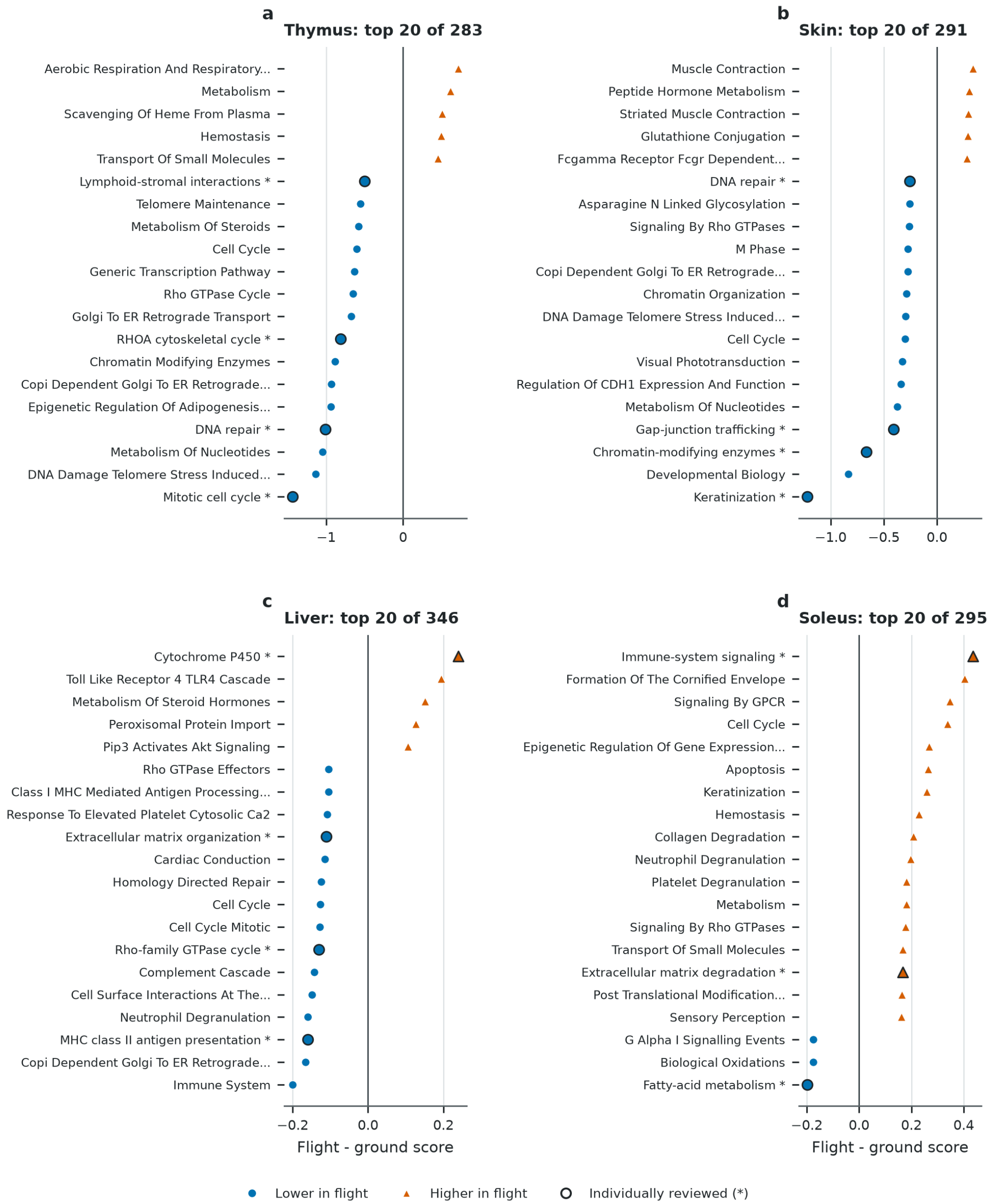


Figure S1. Broad pathway screen for thymus, skin, liver, and soleus. The 20 active Reactome programs with the largest absolute study-balanced effect are displayed for each tissue. Blue circles and orange triangles redundantly encode lower and higher flight scores. Asterisks and black outlines identify pathways selected for individual review. Long labels are shortened for display; full Reactome terms are in Table S9. Ranking is based on magnitude only; Figure S3 and Tables S10-S11 provide the pathway-family and tissue-context review. Kidney and spleen screens are reported in Tables S27-S29.

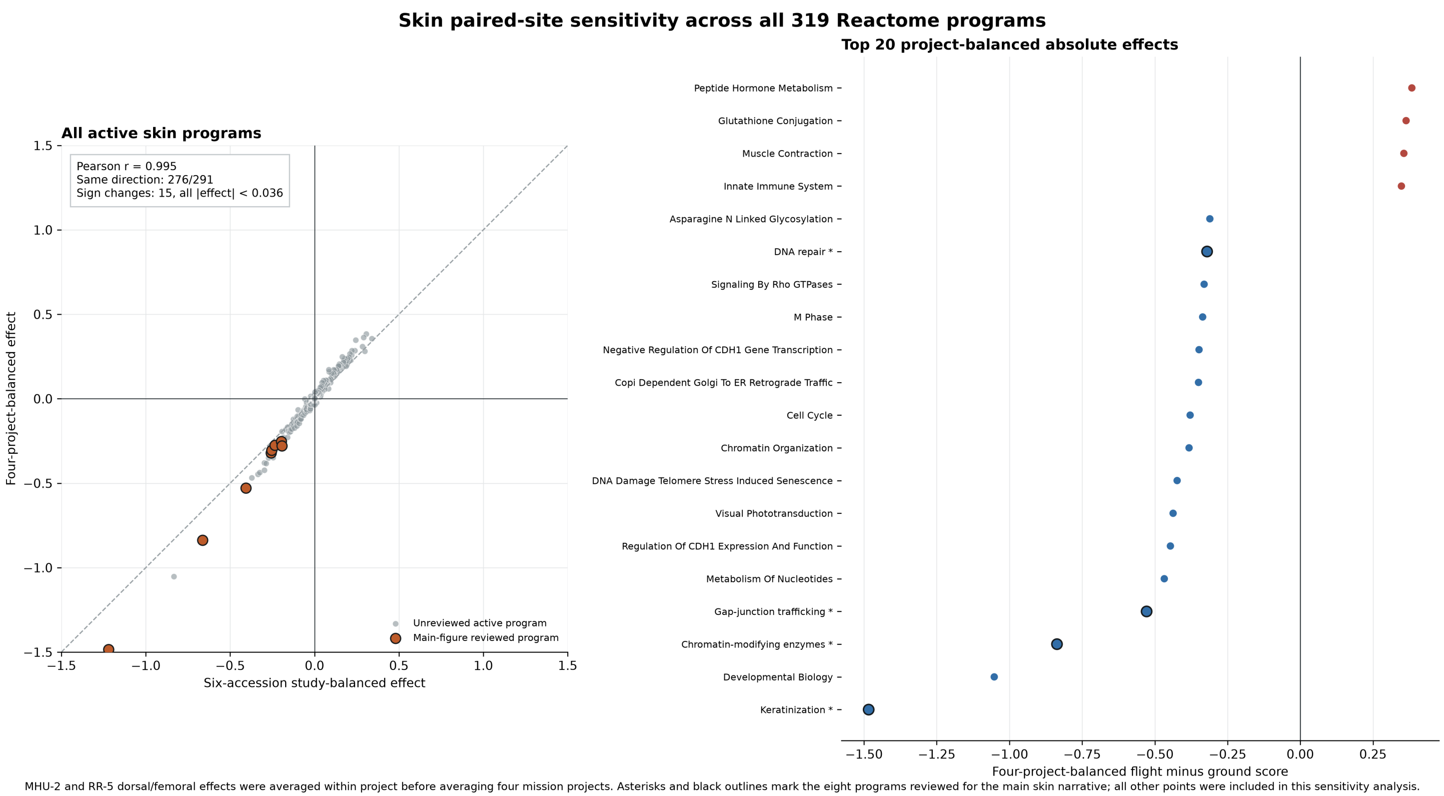
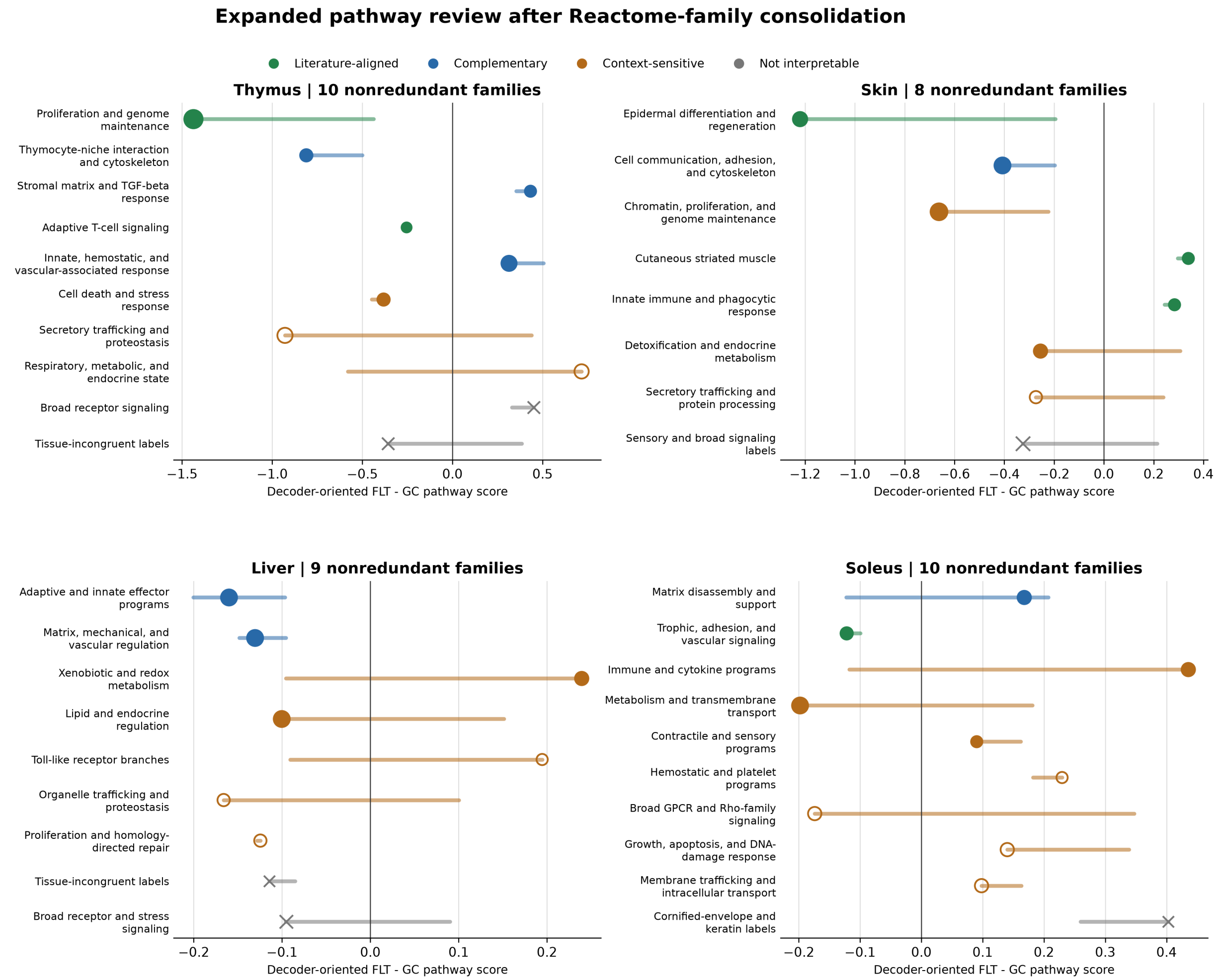


Figure S2. Skin paired-site project-balance sensitivity across the full pathway model. The left panel compares the primary six-accession effect with a four-project effect after averaging paired dorsal and femoral accessions within MHU-2 and RR-5. All 291 active skin pathways are shown. The right panel displays the 20 largest absolute four-project effects. Asterisks and outlined points identify the eight pathways individually reviewed for the main figures; expanded candidate assessments are provided in Figure S3 and Tables S10-S11.



Dots show a literature-reviewed representative; horizontal segments span candidate pathways in the same family. Dot area reflects family size. Hollow points are supplementary-only; x marks excluded broad or tissue-incongruent families. Scales are tissue-specific.

Figure S3. Expanded pathway review after Reactome-family consolidation. For thymus, skin, liver, and soleus, the review included the top within-tissue decile, directionally stable programs through rank 40, and every displayed pathway, yielding 153 pathway records and 37 tissue-specific process families. Dots show a reviewed representative, and segments span candidate effects within each family. Crosses denote broad or tissue-incongruent families excluded from interpretation. Kidney and spleen top-decile reviews are reported separately in Table S28.

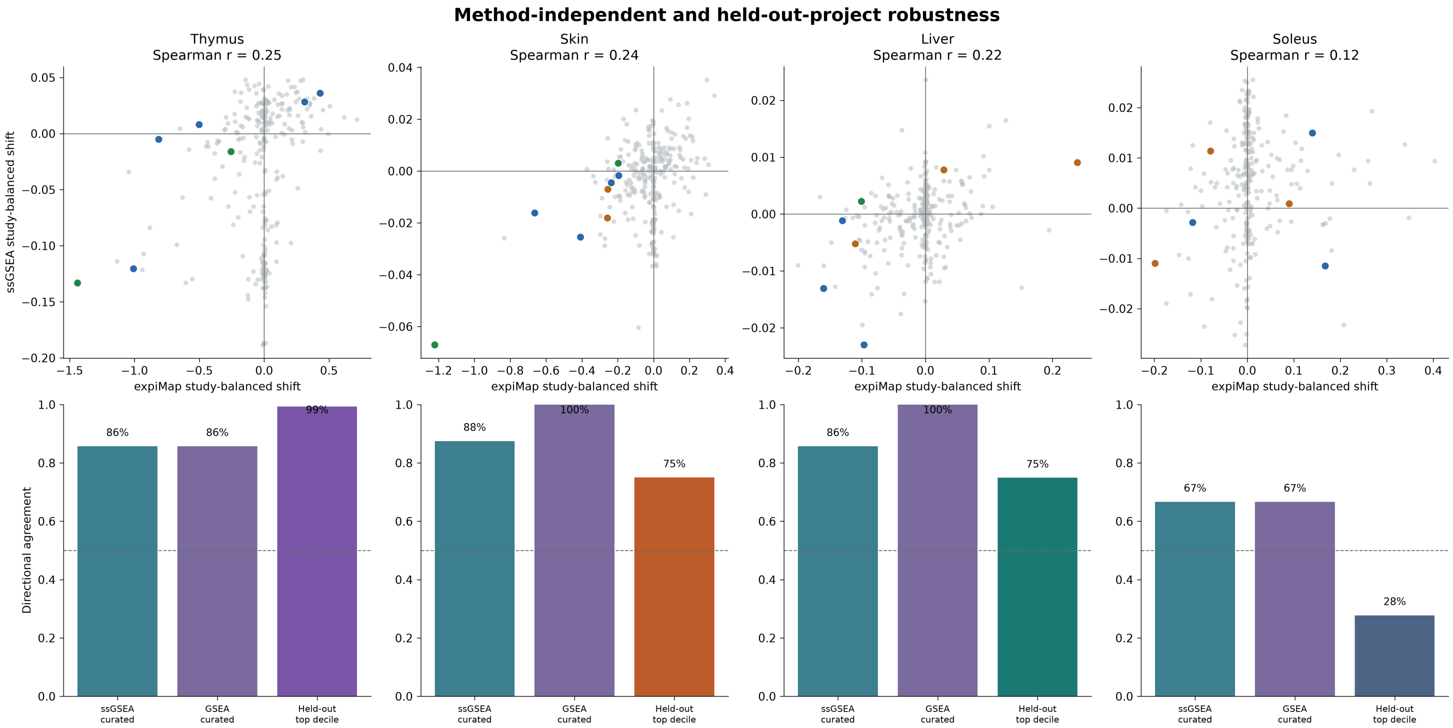


Figure S4. Conventional-method and held-out-project robustness. Top panels compare decoder-oriented expiMap shifts with rank-normalized ssGSEA shifts; colored points are the 29 reviewed thymus, skin, liver, and soleus programs. Bottom panels report directional agreement with ssGSEA and preranked GSEA and training-only top-decile prediction of held-out projects. Held-out results are internal cross-validation, not external replication. Equivalent kidney and spleen analyses are summarized in Figure 4 and Table S27.

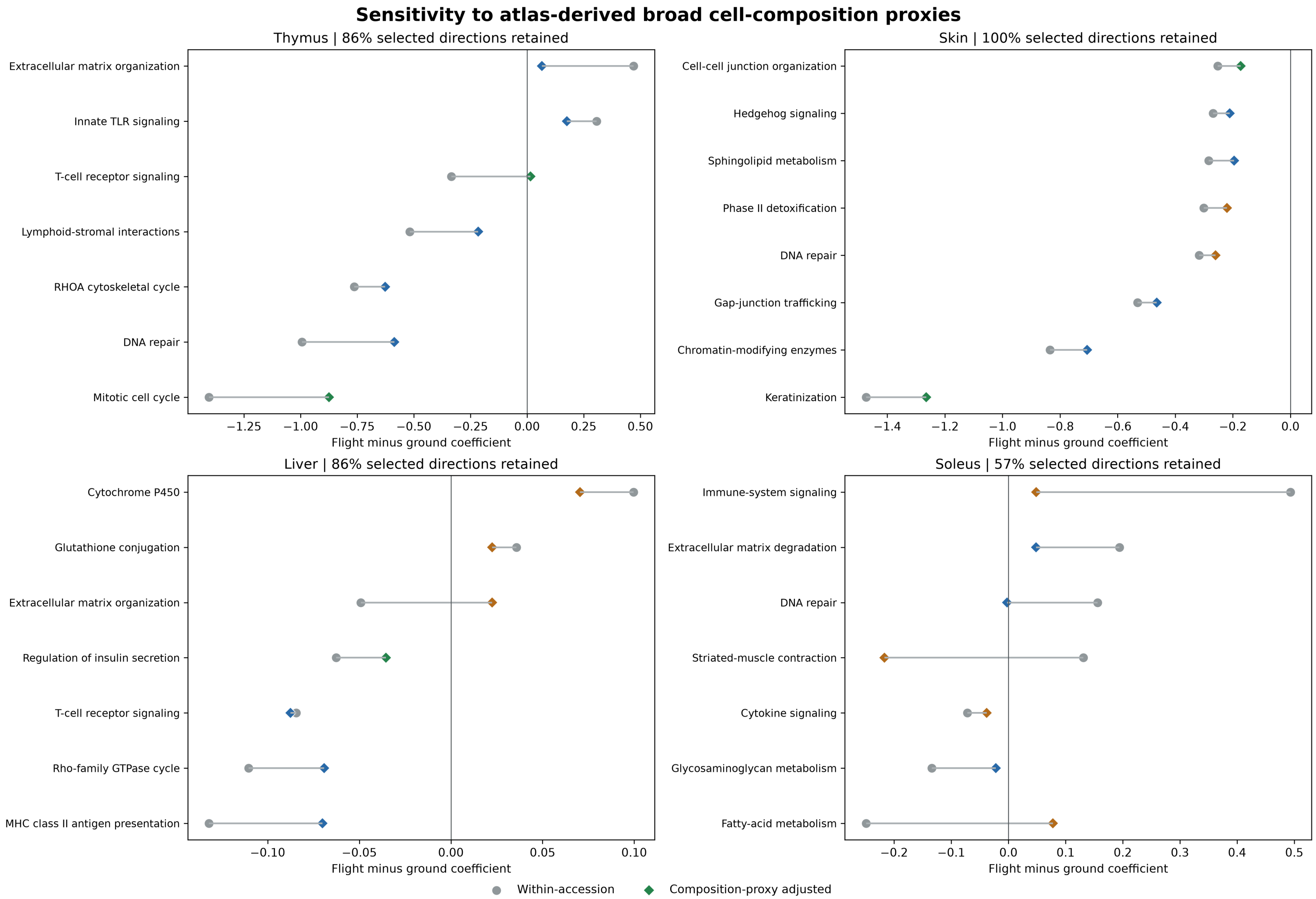


Figure S5. Sensitivity to atlas-derived broad composition proxies. For thymus, skin, liver, and soleus, circles show within-accession flight coefficients before adjustment and diamonds show coefficients after adjustment for up to three principal components of Tabula Muris Senis broad-compartment marker scores. Marker adjustment is a sensitivity analysis, not cell-type deconvolution. Kidney and spleen composition results are reported in Table S27.

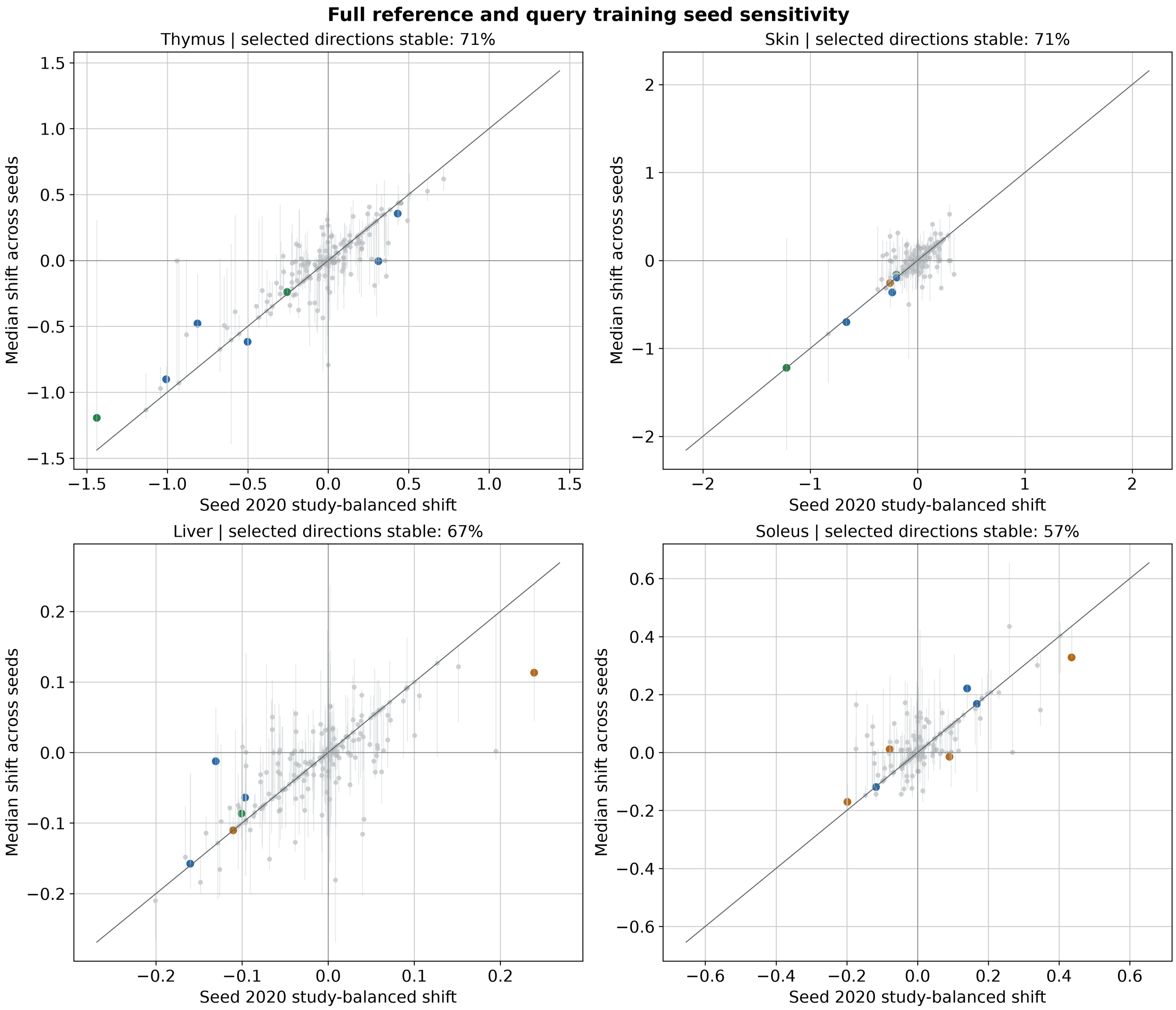


Figure S6. Full reference-query training-seed sensitivity. For thymus, skin, liver, and soleus, the primary seed-2020 effect is compared with the median across three complete reference and query training runs. Segments span the seed range. Unlike query resampling against a fixed reference, this analysis retrains both the ARCHS4 reference and OSDR mapping. Kidney and spleen seed evidence is summarized in Figure 4 and Tables S26-S27.

Original four-model pathway robustness checks

	ssGSEA	GSEA	Held-out	3 seeds	Composition
Thymus: DNA repair	+	+	+	+	+
Thymus: Extracellular matrix organization	+	+	+	+	-
Thymus: Innate TLR signaling	+	+	+	-	+
Thymus: Lymphoid-stromal interactions	-	-	+	+	+
Thymus: Mitotic cell cycle	+	+	+	-	+
Thymus: RHOA cytoskeletal cycle	+	+	+	+	+
Thymus: T-cell receptor signaling	+	+	+	+	-
Skin: Cell-cell junction organization	-	+	+	+	+
Skin: Chromatin-modifying enzymes	+	+	+	+	+
Skin: DNA repair	+	+	+	+	+
Skin: Gap-junction trafficking	+	+	+	-	+
Skin: Hedgehog signaling	+	+	+	+	+
Skin: Keratinization	+	+	+	-	+
Skin: Phase II detoxification	+	+	+	-	+
Skin: Sphingolipid metabolism	+	+	+	+	+
Liver: Cytochrome P450	+	+	-	+	+
Liver: Extracellular matrix organization	+	+	+	+	-
Liver: Glutathione conjugation	+	+	-	-	+
Liver: MHC class II antigen presentation	+	+	+	+	+
Liver: Regulation of insulin secretion	-	+	+	-	+
Liver: Rho-family GTPase cycle	+	+	+	-	+
Liver: T-cell receptor signaling	+	+	+	+	+
Soleus: Cytokine signaling	-	-	+	-	+
Soleus: DNA repair	+	+	+	+	-
Soleus: Extracellular matrix degradation	-	-	+	-	-
Soleus: Fatty-acid metabolism	+	+	-	+	-
Soleus: Glycosaminoglycan metabolism	+	+	+	+	-
Soleus: Immune-system signaling	-	-	-	+	-
Soleus: Striated-muscle contraction	+	+	-	-	-

Figure S7. Pathway-level robustness evidence matrix. Blue cells and plus symbols indicate directional support from ssGSEA, preranked GSEA, held-out projects, all-three-seed concordance, or composition adjustment for 29 reviewed thymus, skin, liver, and soleus pathways; pale cells and minus symbols indicate no support. These are descriptive evidence categories, not statistical significance levels.

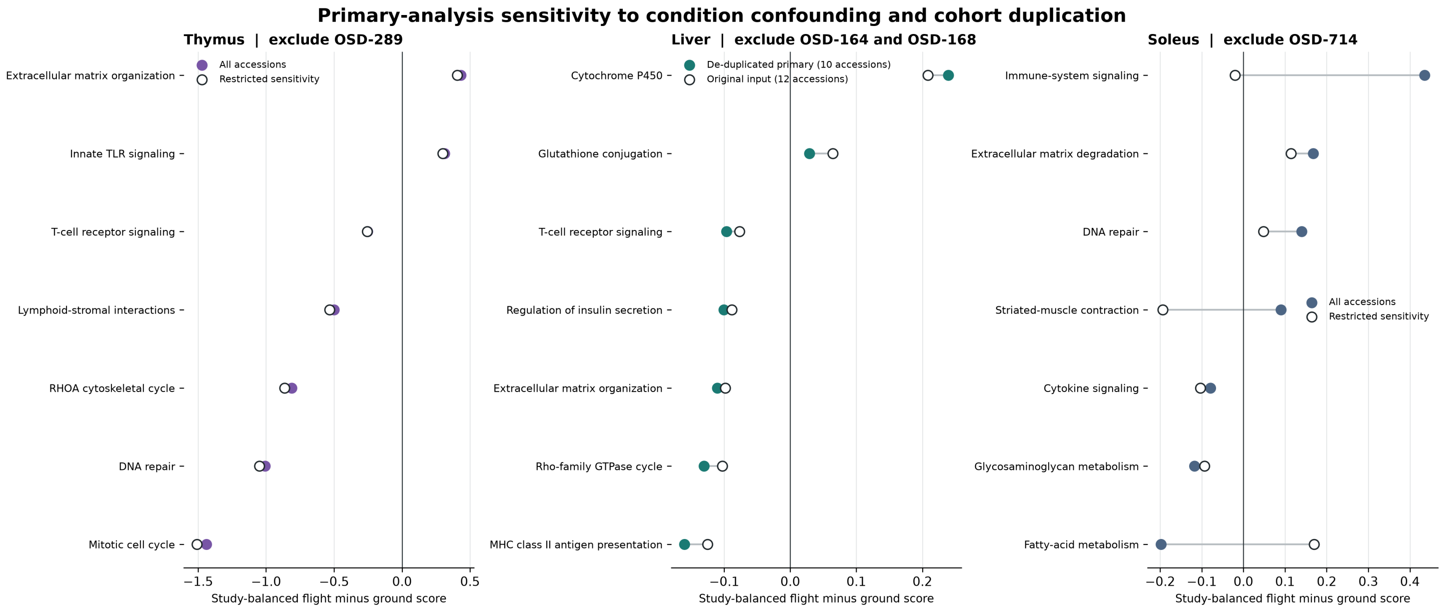


Figure S8. Tissue confounding and cohort-overlap sensitivity. Thymus and soleus effects are compared before and after exclusion of OSD-289 and OSD-714, respectively. Liver compares the de-duplicated 10-accession primary remap with the original 12-accession input containing overlapping cohort representations. Thymus directions were preserved, liver immune directions became more distinct after de-duplication, and several soleus directions changed or attenuated.

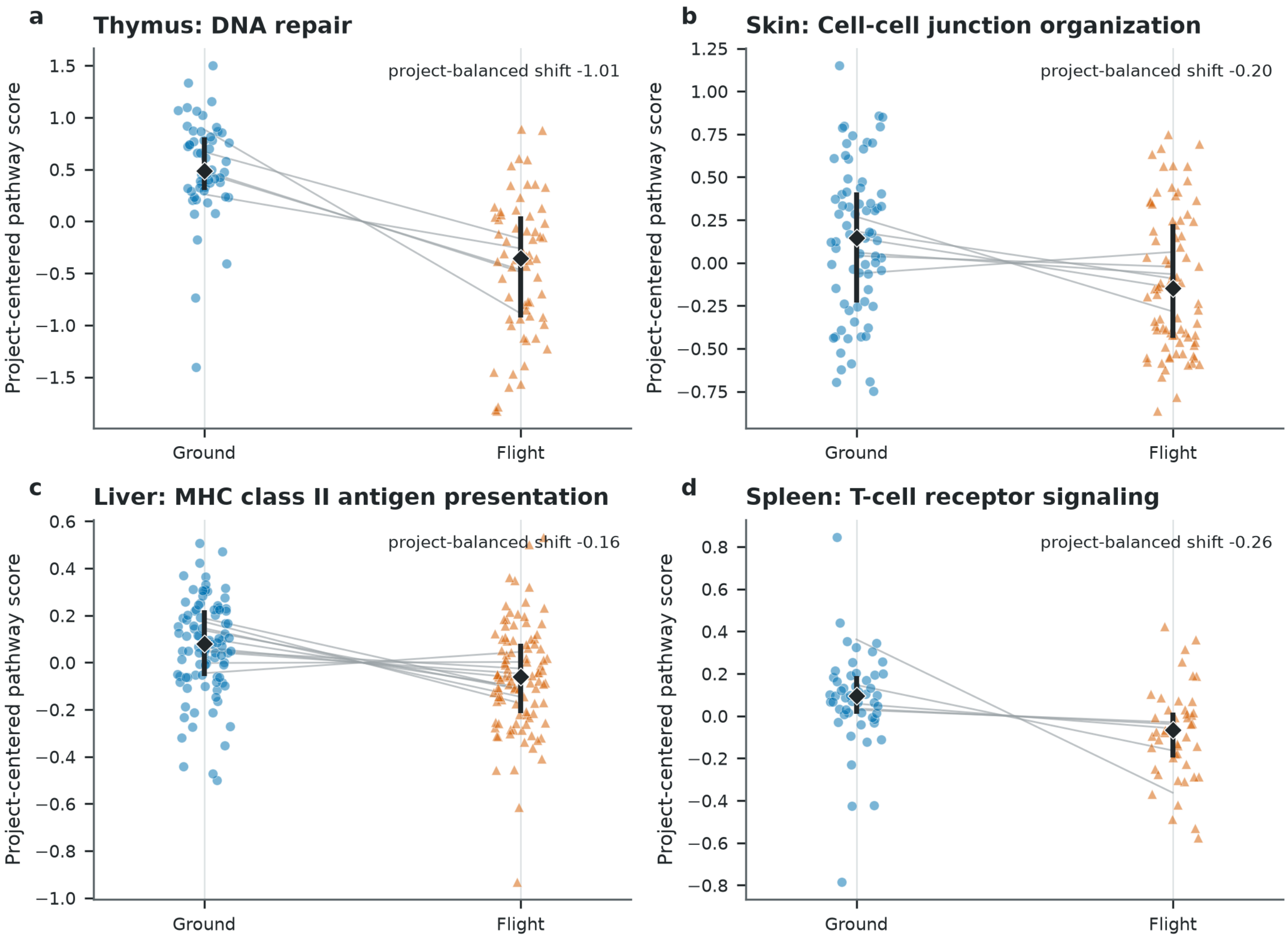


Figure S9. Sample-level distributions for representative retained programs. Individual OSDR samples are shown after decoder orientation and centering within OSD accession. Blue circles are ground controls and orange triangles are flight samples; gray segments connect condition means within a project. Black diamonds and bars show the condition median and interquartile range. The annotated shift is the primary project-balanced expiMap effect, which is not estimated by treating individual points as independent. Skin uses the broad flight label; protocol-defined contrasts are shown in Figure 5.